

From Mileage Proxies to Tread Boundaries: An Auditable Pilot Study of Vision-Based Tyre Inspection

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ABSTRACT Camera-based tyre inspection requires a distinction between recognising collection labels, locating visible boundaries, and measuring physical condition. This paper presents an auditable pilot workflow covering these tasks without treating them as interchangeable. The collection contains 418 unique photographs from 12 operator-confirmed distinct tyres and three ordered mileage-proxy classes. We analyse 153 retained classification runs, explanation-based screening, controlled training interventions, and 81 manually supervised localisation runs. Classification results depend strongly on checkpoint policy: MobileNetV4 obtains mean macro-F1 values of 0.99746 at validation-selected checkpoints and 0.91310 at the fixed-final endpoint. Two historical folds retain identity-overlap concerns. Despite high region overlap, predicted crops and fixed equal-probability fusion do not improve the frozen classifier on average; combining full, tyre, and tread views changes macro-F1 by -0.01324. A separate matched experiment uses 120 annotated images, a fixed 72/24/24 split, and three training seeds. A high-resolution point model reduces mean horizontal boundary error from 1.721% to 1.312% of image width relative to segmentation-derived points, with complete comparator coverage. This 23.75% relative reduction concerns only two test tyres and differing supervision. A native workstation preserves analysed frames, displays geometric evidence and rejection states, and exposes failures on unlabelled videos. The results support an inspectable research workflow, while external generalisation, physical wear estimation, and real alignment accuracy remain unvalidated.

INDEX TERMS Boundary localisation, explainable artificial intelligence, image segmentation, mileage proxy, reproducibility, tyre inspection.

I. INTRODUCTION

VISION-based tyre inspection is attractive because a photograph can preserve the visible condition of a tyre and support repeatable review. However, the meaning of a model output depends on the target used to train it. Recognising a mileage category is different from measuring tread depth; locating a tread boundary is different from recovering a wheel plane; and displaying an angle is different from validating that angle against a physical reference. These distinctions become especially important when a large number of network runs is supported by a small collection of tyres.

This study investigates an evidence-preserving workflow for such a setting. The available collection contains three ordered mileage-proxy labels, manual tyre/tread masks, and

a later six-point boundary annotation set. It does not contain measured tread depths or independently measured alignment angles. We therefore evaluate classification against the recorded proxy labels, localisation against the manual references, and boundary learning against fixed image-space coordinates. We treat the inspection workstation as an implemented integration artifact whose software behaviour can be tested separately from prediction accuracy.

A practical motivation is to test whether adding apparently useful components improves the intended output. A region model may agree closely with a manual mask yet fail to improve classification after cropping. A dedicated point model may improve still-image localisation yet produce crossed boundaries on a new video. Reporting these outcomes to-

gether is more informative than presenting the largest internal score as overall system accuracy. The general problem of leakage and mismatched generalisation claims in machine-learning research further motivates explicit sampling units, split descriptions, and reproducible provenance [1].

The contribution is an empirical pilot and inspectable implementation, rather than a new backbone or a validated physical instrument. Specifically, we provide: (1) an endpoint-aware comparison of 17 retained classification configurations, with identity-based quarantine and explanation-guided follow-ups; (2) manually supervised localisation and paired crop/fusion evaluations that preserve negative results; (3) a same-split, fixed-budget comparison of point learning and segmentation-derived boundaries; and (4) a native workstation that preserves the analysed frame and separates model proposals, supported image geometry, and target-assisted measurement.

The main research questions are whether classification conclusions change with checkpoint endpoint; whether localisation improves a frozen full-image classifier; whether a dedicated point model reduces horizontal boundary error under the matched protocol; and whether these components can be integrated without concealing invalid geometry or domain failures. The historical explanation hypotheses are retained as diagnostics rather than rewritten after observing their outcomes. Figure 1 summarises the evidence flow and its physical-validation boundary.

II. RELATED WORK

A. PHYSICAL MEASUREMENT AND PROXY RECOGNITION

TireEye uses optical observations of longitudinal groove cross-sections, image edges, and tread-wear indicators as a scale reference [2]. Its relevance here is the distinction between a physically referenced measurement and a classification target. Our mileage labels do not supply the scale or reference measurements needed for an equivalent depth estimate. Consequently, we do not compare our macro-F1 to a millimetre error or claim a physical-wear improvement from a classification score.

Residual networks provide a standard recognition baseline [3], while MobileNetV4 targets efficient architectures across mobile hardware [4]. We use available implementations and pretrained representations as experimental configurations, not as evidence that one architecture family is universally appropriate for tyres. For models with ordinal heads, CORAL provides a threshold-based formulation for ordered labels [5]. Its decision rule must be preserved when reconstructing predictions: counting positive ordinal thresholds is not generally interchangeable with taking the largest displayed class score.

B. EXPLANATION AND DENSE LOCALISATION

Grad-CAM uses class-related gradients and feature maps to construct coarse localisation maps [6]. Such maps can help identify where a model responds, but visual plausibility alone is insufficient. Parameter-randomisation sanity checks demonstrate why a saliency method must depend

meaningfully on learned parameters before it is used as model evidence [7]. We combine a project-specific randomisation screen with insertion/deletion diagnostics and explicitly limit the interpretation of spatial saliency ratios.

U-Net, DeepLabV3+, and SegFormer represent different dense-prediction designs [8]–[10]. Our dense-task branch also uses YOLO26 software and a specified RT-DETRv2-R18 model [11], [12]. These configurations differ in pretraining, losses, and backend behaviour. A common image set and epoch budget make their outputs comparable within the declared protocol, but do not create a controlled architecture-only experiment. We distinguish the official research papers from mutable software documentation and preserve executed package versions separately.

HRNet maintains high-resolution representations through parallel multiresolution processing [13]. This motivates its use as a boundary-coordinate component, but human-pose benchmarks do not establish tyre performance. Our six fixed-row coordinates, coordinate decoder, and error definition are task-specific. The matched comparator is a newly trained SegFormer on the same geometry split, not a segmentation checkpoint trained on a broader image allocation.

C. CONFIDENCE AND GEOMETRIC REFERENCE FRAMES

Temperature scaling addresses confidence calibration without requiring a new recognition backbone [14]. Conformal prediction provides a framework for prediction sets under its statistical assumptions [15]; coverage observed on a small, dependent collection does not establish deployment coverage. We therefore report achieved coverage, set size, and the limitations of the inherited abstention field.

ChArUco board detection and pose estimation provide the software foundations for the prototype's target-assisted alignment workflow [16]. A detected target can define a reference plane only when camera calibration, board scale, orientation, and mounting are appropriate. These assumptions are distinct from segmentation quality. The present contribution integrates the software workflow and exercises synthetic cases; it does not supply the missing real-wheel reference study.

III. DATA AND EXPERIMENTAL DESIGN

A. COLLECTION, LABELS, AND SAMPLING UNIT

The preparation audit starts with 888 raw files and removes 470 byte-identical duplicates, leaving 418 clean RGB photographs at 1152×1536 pixels. Ten pre-generated derivatives per original produce 4,180 additional files at 768×768 pixels. These derivatives are linked to their source images and are not independent observations. The photographs were captured within approximately 22 minutes on August 25, 2026, limiting variation in acquisition conditions.

The low-, mid-, and high-mileage proxy groups contain 169, 97, and 152 clean images, respectively, distributed across 3, 3, and 6 timestamp-derived sessions. Low includes new tyres and the 100–5,000 mileage group; mid includes the 40,000, 70,000, and 90,000 groups; high corresponds to the

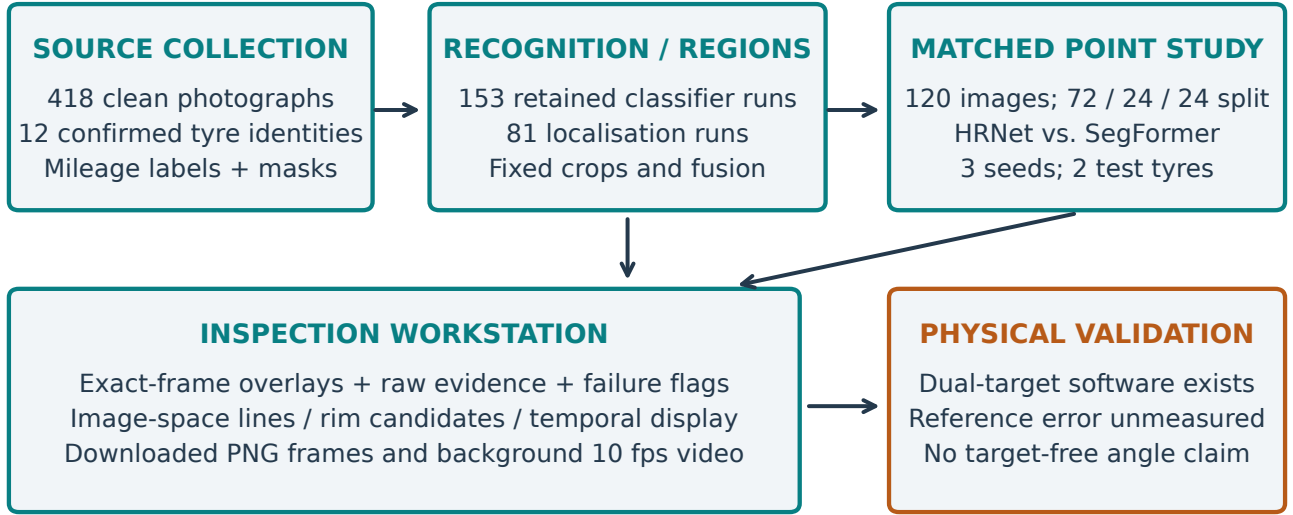


FIGURE 1. Study and workstation overview. The diagram separates completed empirical and software components from physical validation still requiring real reference measurements. The matched geometry study reuses a subset of the source cohort; arrows do not imply independent data collections.

TABLE 1. Historical clean-image allocations. Validation images are reused for internal evaluation; these folds are not an external test set.

Fold	Train	Validation	Interpretation
0	232	186	Historical identity-overlap concern
1	290	128	No listed overlap flag; small internal split
2	314	104	Historical identity-overlap concern

100,000-plus group. These are source-group labels, not measured wear severity. The collection operator subsequently confirmed that the 12 sessions represent 12 distinct physical tyres. This statement is recorded as operator confirmation, without an independent identity audit.

Table 1 preserves the existing folds. Folds 0 and 2 carry historical suspected same-tyre overlap flags. The later operator clarification does not constitute an independent re-audit of these allocations, so the flags remain visible. Classification training can use derivatives of training originals; validation uses clean held-out photographs. The dense-task branch uses clean originals only. Results pooled over folds are descriptive internal summaries. Repeating training with three seeds characterises optimisation variation conditional on the same data and does not triple the number of tyres.

B. MANUAL REGIONS AND GEOMETRY ANNOTATIONS

The existing manual label map M defines overlapping binary regions:

$$Y_{\text{tyre}} = \mathbb{I}[M > 0], \quad Y_{\text{tread}} = \mathbb{I}[M \in \{2, 3\}]. \quad (1)$$

The tread lies within the tyre; treating these labels as mutually exclusive softmax classes would change the task. Bounding boxes are derived from each region. Model agreement with these masks is not a measurement of inter-annotator consistency: no blind independent repeat-annotation study is available.

The geometry extension contains 120 photographs with 720 visible boundary targets. Each image has a left and right tread coordinate on each of three guide rows at 25%, 50%, and 75% of image height. These targets encode horizontal location at a fixed row, not arbitrary landmarks or three-dimensional orientation. The original 12 pilot photographs are excluded from the 120-image package, but the cohort had already informed development. A frozen allocation uses 72 images from eight tyres for training, 24 from two tyres for validation, and 24 from two tyres for testing. Test identities are `mileage_100000_plus` session 006 and `new_tire` session 001. This geometry split is separate from the historical classification folds.

C. ENDPOINTS AND ANALYSIS UNITS

For classification, we preserve both validation-selected and fixed-final checkpoint statistics. They answer different questions, and selection on the evaluated validation data prevents the larger selected score from being treated as an untouched test result. Localisation and the matched geometry comparison use a fixed epoch-60 endpoint. Unless stated otherwise, numerical differences are descriptive; no population significance claim is inferred from repeated seeds, correlated points, or reused image sets. Missing metrics for a box-only model are undefined rather than zero.

IV. METHODS

A. CLASSIFICATION, IDENTITY AUDIT, AND EXPLANATION SCREENING

The retained classification sweep contains 17 configurations, three folds, and three seeds: 153 runs. Nine additional runs labelled ConvNeXtV2-Small are quarantined because their recorded parameter count and a sampled checkpoint's tensor signature identify a ResNet-18 substitution. These records remain preserved but are excluded from the architecture comparison. Model names alone are therefore insufficient provenance; implementation, initialisation, head, and checkpoint identity are part of the experimental specification.

Macro-F1 averages the class-specific F1 values over the three proxy classes. For an ordinal model with threshold outputs $q_k(x)$, the implemented prediction is

$$\hat{y}(x) = \sum_{k=0}^1 \mathbb{I}[q_k(x) > 0.5]. \quad (2)$$

Models trained with a softmax head retain their argmax decision rule. Saved predictions record which rule applies; downstream reconstruction does not replace the ordinal decision with a class-score argmax.

The common classification recipe uses full-frame raw RGB inputs, pretrained full-backbone fine-tuning, a CORAL binary-cross-entropy head, session-balanced sampling, AdamW with initial learning rate 3×10^{-4} and weight decay 0.05, five warm-up epochs, cosine scheduling, and a 60-epoch budget without early stopping. Gradient clipping is 5.0 and training uses half precision. Resolution and batch size follow the architecture registry rather than an identical-resolution assumption: most inputs are 384 pixels, Swin and CoAtNet use 224, and DINOv2 uses 392. Batch sizes range from 16 to 64. Exact per-run configurations and the dataset augmentation policy remain part of the reproducible record.

The explanation screen covers 18 initial candidates and ten shortlist confirmation runs, with 1,208 explanation rows and 35 method-faithfulness rows. A method survives the implemented gate if its randomisation sensitivity exceeds 0.05 and its insertion-minus-deletion area-under-curve value is non-missing. Surviving methods are ranked by that difference. There is no additional requirement that the difference be positive, and passing this gate is not a general certificate of explanation validity.

For a nonnegative saliency map A and tread region T , the normalised tread evidence ratio is

$$\text{TER}_{\text{norm}} = \frac{\sum_{p \in T} A_p / \sum_{p \in \Omega} A_p}{|T|/|\Omega|}, \quad (3)$$

where Ω is the image domain and the ratio is defined only for nonzero denominators. A value above one means concentration beyond the tread's area share. It does not demonstrate causal use of wear features. In this collection, the documented median tread-to-tyre area ratio is 0.990, and 114 images have no visible shoulder. Thus the metric mainly distinguishes tyre from background rather than fine tread from shoulder. Invalid or zero saliency maps remain invalid observations.

The screen selects RegNetY-016, DenseNet-121, and ResNet-50 for one-factor-at-a-time follow-ups. There are 108 discovery runs: three architectures, twelve factor changes, and three seeds on fold 1. A same-fold confirmation stage then evaluates class-weighted sampling, uniform sampling, and random initialisation on ConvNeXtV2-Tiny and MobileNetV4, with three seeds and 60 epochs, giving 18 runs. This tests transfer of factor direction across architectures on the same fold, not independent replication or interactions between separately changed factors.

B. DENSE LOCALISATION AND FROZEN-CLASSIFIER INTERVENTIONS

The localisation branch comprises nine configurations: U-Net/ResNet-34, DeepLabV3+/ResNet-34, SegFormer-B0, SegFormer-B2, YOLO26-n and YOLO26-s in detection and instance-segmentation modes, and RT-DETRv2-R18. Three folds and three seeds produce 81 runs, each trained for 60 epochs on 512-pixel inputs. The declared optimiser is AdamW with learning rate 10^{-4} , weight decay 0.01, and cosine scheduling. Batch sizes are four for semantic and YOLO models and two for RT-DETR.

Semantic models use binary cross-entropy plus soft Dice on two sigmoid channels consistent with (1); the other backends retain their native task losses. Horizontal flipping has probability 0.5 for semantic and repaired YOLO training, while RT-DETR has no online augmentation. YOLO's amended policy disables other hidden augmentations. Final native exponential-moving-average weights are evaluated for YOLO, and final raw weights for the other backends. Differences in pretraining, loss, and backend behaviour are retained as experimental qualifications.

YOLO polygon conversion is verified against the native manual regions. All 836 exported regions meet the required rasterised intersection-over-union (IoU) of at least 0.98; the minimum is approximately 0.980092. This tests the representation conversion, not annotation accuracy. Native masks remain the evaluation reference. Box and mask average precision (AP) use IoU thresholds from 0.50 to 0.95. Region evaluation additionally reports IoU, Dice, and boundary F1 at a two-native-pixel tolerance.

To test downstream usefulness, each localisation run is paired with the frozen final Stage-A ResNet-50 checkpoint of the same fold and seed. Five inputs are evaluated: the full frame, predicted tyre crop, predicted tread crop, oracle tyre crop, and oracle tread crop. Crops include 5% padding. A missing predicted region causes a recorded full-frame fallback. The classifier is not retrained on crops; therefore changes can reflect both localisation errors and input-distribution shift.

The fusion analysis uses saved probabilities rather than new training. Five fixed arms are evaluated: full frame, tyre only, tread only, equal tyre/tread fusion, and equal full/tyre/tread fusion. For the last arm,

$$\bar{p} = \frac{1}{3}(p_{\text{full}} + p_{\text{tyre}} + p_{\text{tread}}). \quad (4)$$

Class decisions are reconstructed with the saved head semantics. Each arm is paired with the corresponding full-frame result. There are 405 run/arm metric rows and 56,430 image/arm prediction rows. These counts are not numbers of independent tyres or independently trained fusion systems. Fusion weights are fixed rather than selected for a favourable test result.

C. CALIBRATION AND PREDICTION SETS

The inherited confidence analysis averages the saved probabilities from the three explanation-selected architectures and their three seeds within each fold. Each fold's held-out photographs are stratified into three disjoint image subsets for temperature fitting, conformal calibration, and final diagnostics, using the recorded seed 100 + fold. This subdivision is image-level, not a new tyre-disjoint split. Temperature is fitted by minimising negative log-likelihood on the first subset. Expected calibration error uses 15 equal-width confidence bins. Unlike the standalone ordinal decision in (2), this inherited ensemble diagnostic explicitly uses the argmax of its averaged class probabilities.

Conformal nonconformity is $1 - p_y$ on the separate calibration subset. The implementation uses the higher empirical quantile at $\min\{1, \lceil (n_{cal} + 1)0.9 \rceil / n_{cal}\}$ and includes test class k when $1 - p_k$ does not exceed that threshold. Coverage and mean set size are then measured on the third subset. The recorded abstention rate counts sets larger than one, leaving empty sets outside that field. We preserve this implementation detail rather than interpreting the field as a complete reject-option metric.

D. MATCHED BOUNDARY LEARNING

The HRNet adapter projects its four feature maps to 16 channels each, upsamples them to the highest feature resolution, concatenates them, and produces six coordinate-logit maps. Bilinear sampling at the fixed guide rows yields a horizontal distribution for each point. The target Gaussian has standard deviation 1.5 feature pixels; training minimises mean cross-entropy with the normalised target distribution. Decoding takes the softmax-weighted expected horizontal coordinate, scaled back to native width minus one. Vertical position is prescribed by the guide, not estimated by the network. HRNet-W18 predicts six horizontal coordinates using Gaussian coordinate targets. The matched SegFormer-B0 learns the two overlapping manual mask channels from the same 72 training images; validation/test dense masks do not enter its training loader. Both use input height/width 512/384, batch size two, AdamW with learning rate 10^{-4} and weight decay 0.01, cosine epoch scheduling, no augmentation, frozen batch-normalisation running statistics, three seeds, and epoch 60 for final comparison. Parameter counts are 9,603,962 and 3,714,658, respectively. The supervision and pretrained backbones differ, so a matched split and budget do not isolate architecture as a causal factor.

SegFormer logits are bilinearly interpolated to native resolution and thresholded at sigmoid 0.5. Tread-row extrema

supply left/right points; empty rows and frame-clipped extrema are rejected. The declared full-set evaluation replaces a missing point with a training-only mean coordinate and separately reports raw coverage and conditional error. This prevents silent evaluation on an easier detected subset. No replacement is needed in the final comparison.

For image i with native width W_i , target j , and training seed s , the horizontal error is

$$e_{ijs} = \frac{|\hat{x}_{ijs} - x_{ij}|}{W_i - 1}, \quad E_s = \frac{1}{6N} \sum_{i=1}^N \sum_{j=1}^6 e_{ijs}, \quad (5)$$

where $N = 24$ test images. The reported percentage is $100E_s$. Each seed contributes 144 target points, giving 432 paired records across seeds on the same two tyres. Relative error reduction uses the SegFormer mean as denominator, not the number of correctly classified images. The mean over seeds is not an evaluated ensemble.

E. INSPECTABLE GEOMETRY AND TARGET-ASSISTED ALIGNMENT

The workstation applies the deployed models to the exact displayed analysis frame. Orientation follows the video metadata and the established portrait policy. HRNet and matched SegFormer use seed 1, final epoch 60, with full-image re-sizing, ImageNet normalisation, and no test-selected crop or rotation search. Local inference uses full precision; the saved training evaluation used automatic mixed precision. Adapter checks therefore verify coordinate/preprocessing consistency without claiming a new aggregate benchmark reproduction.

Image geometry is fitted to observed image evidence rather than obtained by translating class scores into angles. A segmentation mask guides the search region. The boundary path samples Canny edges near the two silhouette sides and uses deterministic consensus followed by least-squares fits of $x = ay + b$. Accepted supports must be sufficiently long and consistent; disagreeing sides, collapsed fits, fragmented masks, and clipping can withhold the fit. A separate side-view mode fits an inner ellipse candidate only when image-edge support and angular coverage are sufficient. A tyre silhouette is not automatically called a rim. Accepted side lines yield an image-space midpoint line, not camber or toe.

Temporal display tracking smooths consistent proposals and resets on discontinuities, source changes, or invalid results. Raw learned coordinates remain available, and crossed or missing boundaries receive review flags. A stable display is not treated as evidence of accuracy. The explanation view distinguishes the predicted search outline from accepted edges and explicitly labels a withheld fit.

The physical-measurement path instead requires calibrated camera intrinsics and two separately identified ChArUco targets [16]. The ground target defines the reference frame and the wheel target supplies a candidate wheel-plane normal. With vehicle axes X forward, Y left, and Z up, and outward-

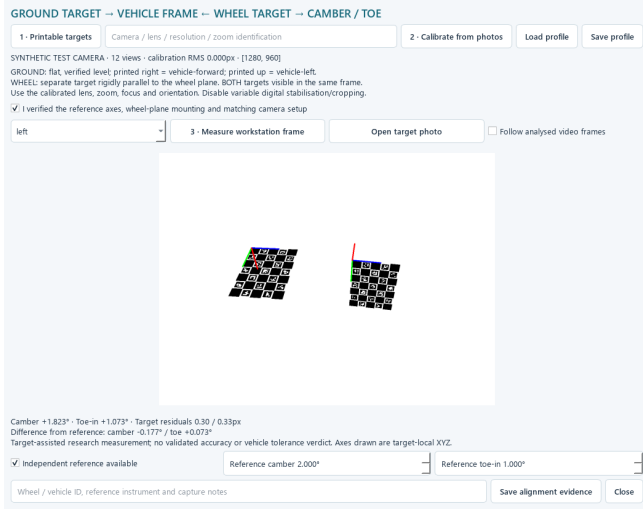


FIGURE 2. Target-assisted alignment interface exercised with a *synthetic test camera*. The displayed angles and residuals are *software-test outputs* from rendered targets, not measured accuracy on a physical wheel.

facing normal n after the wheel-side convention is applied, the implemented angles are

$$\gamma = \text{atan2}(-n_z, \sqrt{n_x^2 + n_y^2}), \quad (6)$$

$$\tau = \text{atan2}(n_x, |n_y|), \quad (7)$$

for camber and toe-in under that explicit setup convention. The workflow requires setup confirmation and rejects missing or unsuitable pose evidence. These equations are software definitions contingent on target orientation and wheel-plane mounting; they do not validate physical accuracy. Real fixture alignment, reference measurements, and remount/runout studies remain necessary.

F. EXECUTION AND REPRODUCIBILITY

Kaggle sessions provide training compute and Hugging Face stores persistent records. Jobs retain model/optimiser state, scheduler and scaler state where applicable, random-number-generator state, and execution position. The older dense engine resumes completed epochs; the later geometry engine checkpoints completed optimiser steps and the next-batch cursor. Normal publication is batched at roughly 30-minute intervals, with major completion and catchable interruption requesting a flush. Remote rate limits, failed uploads, or a forced operating-system kill can still lose unpublished progress.

The geometry engine's deterministic resizing repair passed a Tesla T4 resume check with zero maximum parameter difference. This verifies that exercised interruption path, not every possible hardware failure. Immutable revisions and hashes identify the reviewed artifacts. The present paper regenerates its numeric tables and plots from those frozen local records; it does not rerun training or claim independent replication of the original experiments.

TABLE 2. Classification endpoints on the historical folds. Each row averages nine runs (three folds and three seeds); folds 0 and 2 retain overlap concerns. These are internal validation results, not an independent-test ranking.

Architecture	Selected	Final
MobileNetV4	0.99746	0.91310
Swin-T	0.97673	0.83478
EfficientNetV2-S	0.97615	0.70874
Swin-S	0.95259	0.85744
CLIP ViT-B/16	0.95076	0.78897
ResNet-50	0.93651	0.82922
RegNetY-016	0.93495	0.74087
ViT-S	0.93235	0.75858
ResNeXt-50	0.92260	0.76566
MaxViT-T	0.92189	0.85982
DINOv2-B	0.91277	0.88380
ConvNeXtV2-T	0.90891	0.90238
DenseNet-121	0.89660	0.83852
VGG16-BN	0.89194	0.45870
DINOv2-S	0.87346	0.78842
CoAtNet-0	0.80537	0.72041
DeiT-III-S	0.74726	0.72855

V. RESULTS

A. CLASSIFICATION IS SENSITIVE TO FOLD AND ENDPOINT

Simple baselines reveal substantial fold dependence. The colour probe obtains macro-F1 values of 0.88696, 0.46405, and 0.12322 on folds 0, 1, and 2. The histogram-of-oriented-gradients support-vector machine obtains 0.79602, 0.18182, and 0.98412. These patterns motivate scrutiny of capture-specific cues, but do not identify a unique shortcut mechanism. A historical random-initialised ResNet-18 trained for 15 epochs averages 0.82135, whereas the later matched nine-run, 60-epoch ResNet-50 control averages 0.82332 at the final endpoint. The two controls differ in backbone and budget and are not interchangeable.

Table 2 and Fig. 3 show the retained sweep. MobileNetV4 has a selected mean macro-F1 of 0.99746 and final mean of 0.91310. The endpoint gap is larger for EfficientNetV2-S (0.97615 versus 0.70874) and VGG16-BN (0.89194 versus 0.45870), while ConvNeXtV2-Tiny is comparatively stable (0.90891 versus 0.90238). Consequently, a near-perfect selected score cannot stand alone as a performance summary. These values include the historically flagged folds and do not establish external generalisation, physical-wear recognition, or the optimal checkpoint policy for new tyres.

B. EXPLANATION DIAGNOSTICS AND SAME-FOLD CONFIRMATION

The inherited hypothesis that normalised tread evidence predicts cross-fold stability better than accuracy is not supported: the recorded primary and reference statistics are 0.42710 and 0.90743, respectively, across ten entries. This analysis uses legacy selected-endpoint statistics and is not relabelled as a fixed-budget result. The marking/damage hypothesis is inconclusive or undefined, and the fine-grained architecture

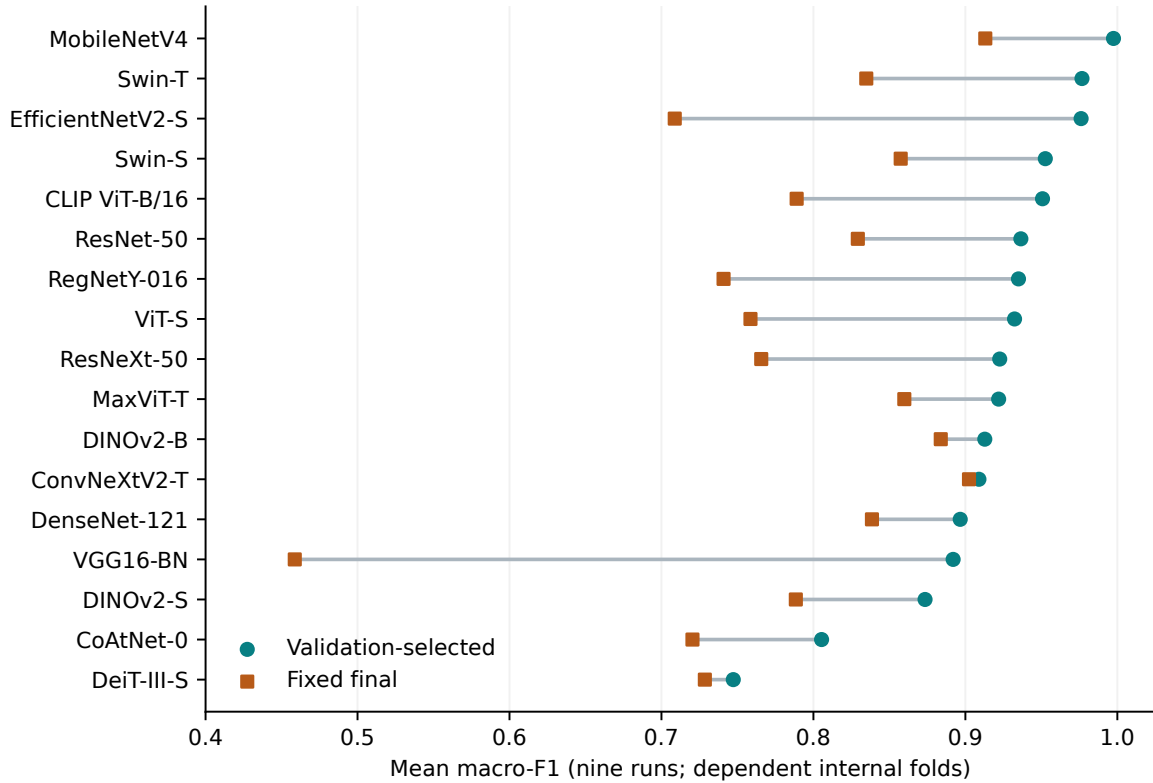


FIGURE 3. Validation-selected and fixed-final macro-F1 means for all 17 retained configurations. Each point averages nine runs on the same historical folds; connecting segments highlight endpoint differences and are not uncertainty intervals. No test-set architecture ranking is implied.

TABLE 3. Same-fold confirmation: mean macro-F1 change relative to each matched baseline over three seeds. Selection and final endpoints can give different sampling conclusions.

Architecture	Factor	Δ selected	Δ final
ConvNeXtV2-T	Class-weighted	+0.01257	+0.02396
ConvNeXtV2-T	Uniform	+0.00000	+0.01027
ConvNeXtV2-T	Random init.	-0.42857	-0.38737
MobileNetV4	Class-weighted	-0.02839	+0.01886
MobileNetV4	Uniform	-0.06959	-0.17079
MobileNetV4	Random init.	-0.37150	-0.27525

hypothesis remains untested because its planned model arms are absent. In particular, fold-1 marking removal operates on zero active marking images; its near-zero effect cannot establish marking independence.

Table 3 reports the confirmation stage. Random initialisation is negative for both architectures at both endpoints. Class-weighted sampling has a positive selected effect for ConvNeXtV2-Tiny but a negative selected effect for MobileNetV4, while both final effects are positive. Uniform sampling is not consistently beneficial. The supported interpretation therefore specifies architecture, endpoint, and the existing fold. The experiment neither isolates factor interactions nor establishes a population-level sampling benefit.

C. REGION OVERLAP AND BOUNDARY FIDELITY DIFFER

All 81 localisation runs complete 60 epochs and evaluation, corresponding to 4,860 epoch records. Table 4 restricts the principal summary to fold 1; this avoids treating flagged-fold averages as the main estimate but does not make fold 1 an external test. Semantic models have tread IoU between approximately 0.972 and 0.976, whereas their two-pixel boundary F1 is approximately 0.358–0.374. YOLO instance-segmentation models obtain tread IoU near 0.856–0.858 and boundary F1 near 0.189–0.226.

Figure 4 makes the metric distinction explicit. Region overlap rewards agreement across a large interior area; boundary F1 is sensitive to contour displacement under its narrow tolerance. Neither metric alone establishes that the model has recovered the information needed for proxy classification, let alone physical tread condition. The evaluation is against one existing manual reference set rather than an independently repeated annotation standard.

D. PREDICTED CROPS AND FIXED FUSION HAVE NEGATIVE MEAN EFFECTS

The paired frozen-classifier analysis does not show an overall gain from the predicted crops. Mean macro-F1 changes are -0.01257 for tyre-only and -0.01467 for tread-only inputs across the 81 localisation source runs. These means do not imply that every configuration/fold pair worsens. They quantify

TABLE 4. Fold-1 localisation means over three seeds. AP denotes COCO-style AP at IoU thresholds 0.50:0.95. Boundary F1 uses a two-native-pixel tolerance. Dashes denote undefined mask metrics for box-only models.

Model	Box AP	Mask AP	Tread IoU	Boundary F1
DeepLabV3+/R34	0.98612	0.99675	0.97176	0.35988
RT-DETRv2-R18	0.94073	—	—	—
SegFormer-B0	0.91189	0.99836	0.97372	0.37258
SegFormer-B2	0.97657	0.99982	0.97569	0.37412
U-Net/R34	0.91413	0.99704	0.97216	0.35820
YOLO26-n det.	0.95278	—	—	—
YOLO26-n seg.	0.95204	0.95890	0.85815	0.18926
YOLO26-s det.	0.96081	—	—	—
YOLO26-s seg.	0.95227	0.96215	0.85627	0.22623

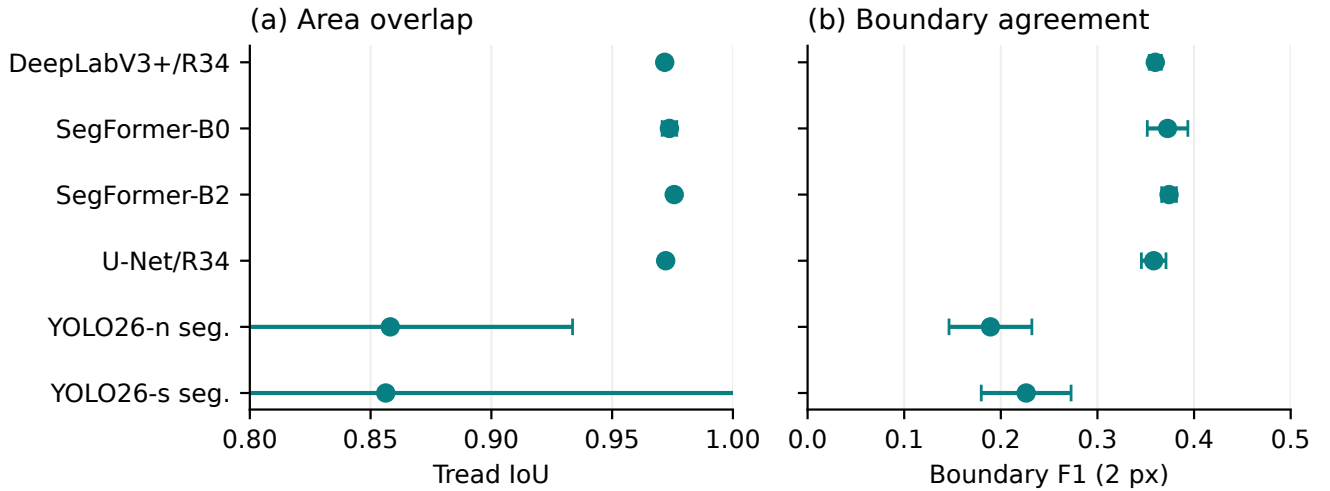


FIGURE 4. Fold-1 semantic/instance-segmentation results: (a) tread IoU and (b) boundary F1 at two native pixels. Points are means and whiskers are sample standard deviations across three training seeds. The panels deliberately use different horizontal scales. Seed variation is not a confidence interval over independent tyres.

the declared intervention on a classifier trained for full-image inputs, with no crop-specific retraining.

Fixed probability fusion also has negative mean effects (Table 5, Fig. 5). Equal tyre/tread fusion gives -0.01568; equal full/tyre/tread fusion gives -0.01324. The latter is a decrease of about 1.324 macro-F1 percentage points, not a 1.324% relative error reduction. The result argues against assuming that localisation or additional views automatically help an existing classifier. Lost context, changed scale, crop distribution shift, localisation errors, and correlated predictions are plausible explanations, but the present design does not separate their contributions. Crop-trained recognition and learned fusion would require new experiments.

E. CONFIDENCE DIAGNOSTICS ARE FOLD-DEPENDENT

Temperature scaling leaves the recorded macro-F1 unchanged and reduces confidence-error metrics on the retained calibration-test subsets (Table 6). On fold 1, expected calibration error decreases from 0.18955 to 0.07377, and negative log-likelihood from 0.35630 to 0.16226, with 41 test images. Extremely small calibrated errors on the flagged folds are not evidence of deployment-grade certainty.

TABLE 5. Fixed probability fusion: mean paired macro-F1 changes over 81 source runs. Repeated images and shared classifiers make these rows dependent.

Arm	Rows	Mean Δ
Full frame	81	+0.00000
Tyre only	81	-0.01257
Tread only	81	-0.01467
Tyre + tread	81	-0.01568
Full + tyre + tread	81	-0.01324

At nominal 90% coverage, the observed conformal coverages are 0.86885, 0.97561, and 0.93939 on folds 0, 1, and 2, respectively (Table 7). Thus only fold 0 is below nominal in these observed subsets. Mean set sizes below one on folds 0 and 2 imply empty sets. The inherited abstention field does not count all empty-set events and cannot be presented as a complete rejection policy. Small subset sizes, dependent acquisition, and reuse of internal evidence constrain generalisation of these diagnostics.

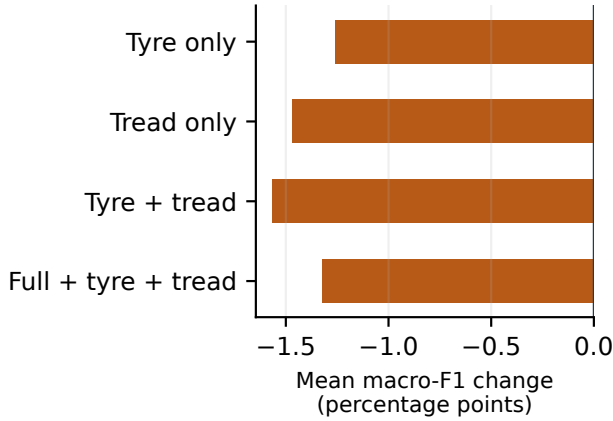


FIGURE 5. Mean paired changes from the full-frame classifier for four fixed crop/fusion arms. Bars average 81 dependent source runs; no independent-sample uncertainty or favourable-arm selection is implied.

TABLE 6. Confidence diagnostics on held-out calibration-test subsets. ECE is the recorded expected calibration error; values near zero on flagged folds do not imply external reliability.

Fold	Scores	n	F1	ECE	NLL
0	Raw	61	1.0000	0.2115	0.2425
0	Temp.	61	1.0000	0.0000	0.0000
1	Raw	41	0.8030	0.1896	0.3563
1	Temp.	41	0.8030	0.0738	0.1623
2	Raw	33	1.0000	0.1293	0.1438
2	Temp.	33	1.0000	0.0000	0.0000

F. MATCHED POINT LEARNING REDUCES STILL-IMAGE ERROR

Table 8 reports the fixed-final comparison, independently recomputed here from the 432 paired point records. Mean horizontal error is 1.312% of native width for HRNet and 1.721% for matched SegFormer, corresponding to approximately 15.10 and 19.80 native pixels. The absolute difference is 0.409 percentage points of width, or 4.70 pixels. Relative to the SegFormer mean, HRNet reduces error by 23.75%. SegFormer raw coverage is 100%, with zero fallback substitutions, so its raw and fallback-assisted means coincide.

HRNet has lower mean error in every seed and on each of the two test tyres (Fig. 6). This does not mean every individual point improves. The new-tyre example has higher error for both models than the high-mileage example, but two tyres cannot establish a class effect. The result supports HRNet as a candidate point-localisation component within this matched study. It does not establish an architecture-only advantage because supervision and initial representations differ, nor does it demonstrate downstream classification improvement or physical-angle accuracy.

G. WORKSTATION INTEGRATION AND OBSERVED VIDEO FAILURES

Tread Station is a native PySide6/PyTorch application with photo/video loading, portrait-aware playback, asynchronous

TABLE 7. Prediction-set diagnostics at nominal 90% coverage. The recorded abstention field does not count every empty-set event.

Fold	n_{cal}	n_{test}	Coverage	Set size	Abstain
0	62	61	0.8689	0.8689	0.0000
1	43	41	0.9756	1.0976	0.0976
2	35	33	0.9394	0.9394	0.0000

TABLE 8. Matched fixed-epoch-60 horizontal point error (% of native width minus one). Each seed uses the same 24 images from two test tyres; coverage is the SegFormer raw point coverage.

Seed	HRNet	SegFormer	Coverage (%)
1	1.391	1.788	100
2	1.284	1.705	100
3	1.261	1.668	100
Mean	1.312	1.721	100

inference, comparison, geometric overlays, and saved evidence restoration. Existing classifier and region checkpoints remain separate from the optional HRNet/matched-SegFormer geometry components. An inference result is attached to its analysed frame; a newer live preview is shown separately rather than receiving a stale mask. Figure 7 shows the implemented interface, and Fig. 8 pairs an operational still example with a preserved video failure.

The saved adapter checks cover strict checkpoint loading, preprocessing, coordinate decoding, repeatability, blank-input flags, temporal resets, and immutable source pixels. A local check on one internal still measured five warm inference calls on an NVIDIA GTX 1650. HRNet's median was 72.60 ms (range 51.41–88.46 ms); matched SegFormer's median was 33.48 ms (33.32–33.84 ms). These timings include pre-processing, inference, extraction, and GPU synchronisation, but exclude model loading and file decoding. Peak allocated memory in the two-component check was approximately 182 MiB. These small operational samples are not a controlled training-hardware comparison, sustained camera throughput, or a guaranteed frame rate.

The supplied videos expose crossed or collapsed HRNet boundaries, missing matched boundaries, and disagreement between models. Crossed pairs withhold widths and connecting geometry while retaining raw proposals for review. No video point ground truth is available, so the passing playback/UI tests establish software behaviour rather than localisation accuracy. Temporal smoothing cannot correct a systematically wrong proposal, and still-image gains cannot be transferred to video by assertion.

The workstation additionally exports the shown analysed frame as PNG and processes an opened video separately at ten output frames per second. The latter runs with a separate CPU engine and frozen display/recipe settings, leaving the GPU available for inspection. The software checks cover frame selection, output duration, portrait dimensions, cancellation, and incomplete-output cleanup; a short real-model portrait excerpt verifies the complete export path. Export frame rate

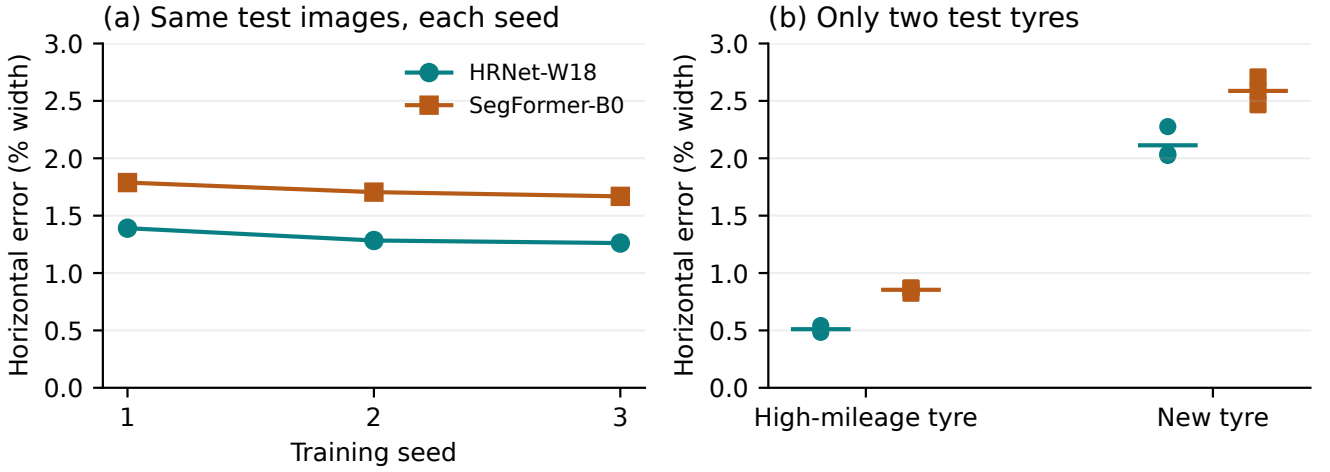


FIGURE 6. Matched boundary comparison. (a) Mean horizontal error for each training seed on the same 24 images. (b) Per-tyre error, retaining all three seed values; horizontal marks are seed means. HRNet is lower in each seed/tyre comparison. These observations are correlated and provide no independent-tyre significance interval.

describes the saved playback, not inference speed. This facility supports evidence sharing and does not add an accuracy result.

H. ALIGNMENT SOFTWARE IS TESTED SYNTHETICALLY

The target-assisted interface generates distinct ground/wheel boards, creates or loads a camera profile, solves target poses, applies the declared side convention, and records signed outputs with pose residuals. It also accepts an independent reference reading for comparison. Synthetic tests exercise known poses, both wheel sides, missing targets, scaling/orientation, setup gating, and evidence export (Fig. 2). The supplied videos do not contain this calibrated dual-target setup. No result here measures real wheel-angle error, fixture mounting error, or runout compensation.

VI. DISCUSSION

A. WHAT THE EVIDENCE ESTABLISHES

The results support three distinct empirical conclusions. First, checkpoint endpoint materially changes the internal classification picture: selected validation performance and fixed-final performance cannot be combined into one accuracy claim. Second, high agreement with a manual region does not guarantee benefit to a full-image classifier after cropping or fixed fusion. Third, dedicated point learning reduces horizontal error in the matched still-image experiment, while integrated video observations reveal limitations outside that evaluation setting.

The engineering contribution is to preserve these distinctions through the workflow. Identity checks prevent a substituted model from entering an architecture comparison. Saved probabilities support paired downstream evaluation without retraining. Explicit coverage prevents a point extractor from reporting only its successful cases. Exact-frame overlays make a displayed prediction auditable, and withholding un-

supported geometry prevents an attractive line from being mistaken for an accepted measurement. These design choices improve traceability; they do not create independent data or remove model error.

The matched point result is useful for component choice, but should be interpreted alongside operational cost and failure visibility. HRNet has more parameters and a higher observed warm inference time in the local check. Conversely, SegFormer provides dense regions that a six-point output cannot replace. The two models can therefore serve complementary roles: dense support for the search region and point proposals for boundary inspection. An evaluated improvement to the full inspection task would require a predefined end-to-end metric and component-removal study; it cannot be inferred by adding component scores.

B. THREATS TO VALIDITY

Construct validity: The classification labels describe source mileage groups, not tread depth, damage, or roadworthiness. Manual regions and fixed-row points are image-space references. There is no independently measured physical-alignment outcome for the current photographs or videos.

Sampling and internal validity: The effective cohort is small and acquisition is concentrated in one brief window. Two historical folds retain identity-overlap concerns, and the unflagged fold remains internal. Operator-confirmed tyre identities improve the documentation but are not an independent audit. Multiple architectures and follow-ups reuse the same data, and selected-checkpoint evaluation reuses validation information. The geometry test contains only two tyres and comes from a cohort already inspected during development.

Comparison validity: The backends have different pretraining, native objectives, parameter counts, and exponential-moving-average behaviour. HRNet receives

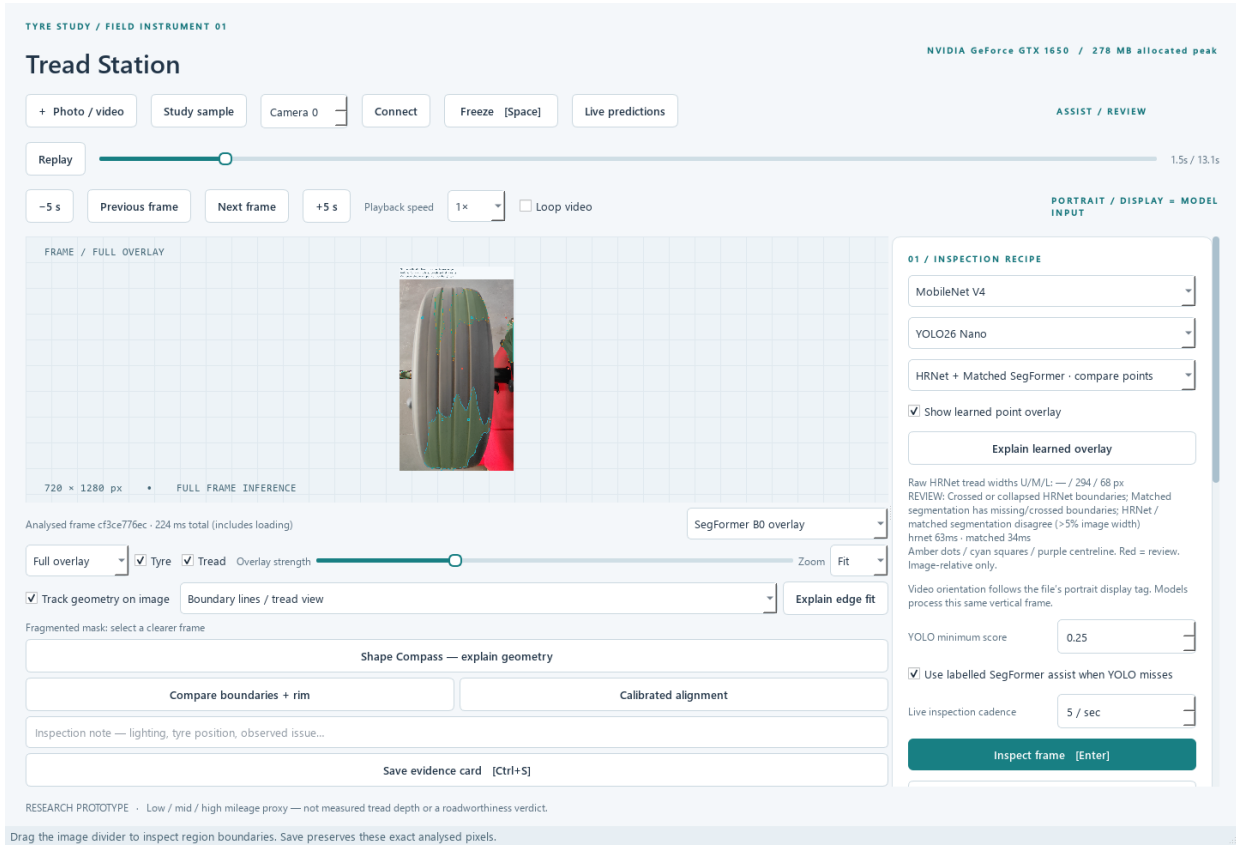
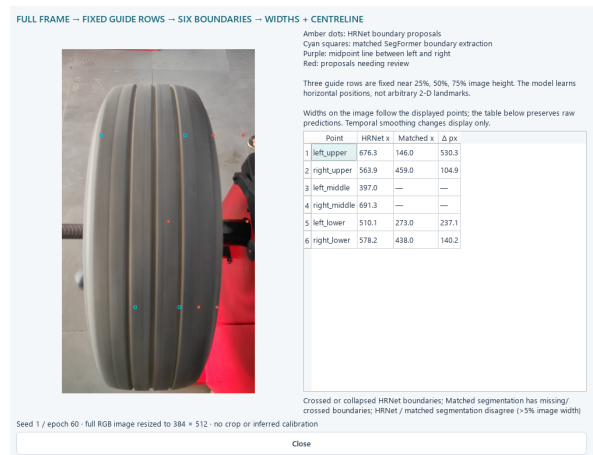


FIGURE 7. Saved native workstation view with learned geometry enabled. This operational screenshot predates the additional download controls. The inspected frame, source preview, model observations, and review flags are distinct. Interface completion is not field-accuracy validation.



(a) Selected internal still-image check



(b) Preserved video-domain failure

FIGURE 8. Qualitative operational evidence, not a new held-out benchmark. Amber HRNet proposals, cyan matched-SegFormer points, and the image-space centreline are shown on the still. The video explanation retains crossed/missing boundaries and disagreement rather than concealing failure. All widths and slopes are image-relative. The images are reproduced from the saved integration checks without altering predicted coordinates.

point supervision and SegFormer receives dense masks. Equal image allocations and budgets do not isolate architecture alone. The S5 region SegFormer and the later matched geometry SegFormer have different training allocations and must not be conflated.

Measurement and conclusion validity: IoU and boundary

F1 emphasise different spatial properties. The saliency ratio is limited by near-coincident tyre/tread regions. Empty conformal sets require explicit handling. Seed standard deviations describe repeated optimisation on fixed data; 432 point records, 405 fusion rows, and hundreds of training runs do not create corresponding numbers of independent subjects.

We therefore avoid population significance or universal superiority claims.

Operational validity: Still-image adapter checks and short video software tests do not establish field robustness, sustained throughput, or physical webcam performance. The observed video failures indicate a domain gap, although their causes cannot be isolated without labelled video evaluation. Recoverability depends on the last successful durable publication and the exercised checkpoint path, not an unconditional guarantee against interruption.

C. IMPLICATIONS FOR PHYSICAL ALIGNMENT AND FUTURE STUDIES

A projected boundary or ellipse can change with viewpoint, lens distortion, occlusion, and tyre shape. Converting its image slope directly into toe or camber would require assumptions not supplied by the current images. The target-assisted workflow provides an explicit route to a reference frame, but only a real calibration/fixture study can assess whether the reference frame corresponds to the wheel plane and vehicle axes with useful accuracy.

The next data collection should prioritise independently identified tyres and varied cameras, illumination, views, and operating conditions. For boundary learning, a tyre-disjoint video test should be frozen before tuning and should measure point error, missing/crossed-boundary rates, and temporal behaviour. For recognition, crop-trained models and learned fusion should be evaluated as new hypotheses under a selection rule that does not reuse the final test set. Physical studies should include calibrated reference captures, independently measured wheel angles, repeated target mounting, and runout assessment. A healthy-reference anomaly branch also requires justified healthy labels; low mileage alone is insufficient. These are proposed extensions rather than completed results.

VII. ARTIFACT AVAILABILITY AND REPRODUCIBILITY

Source code and maintained documentation are available at <https://github.com/Shanmuk4622/tyre-wear-alignment-cv>. The study's persistent artifact repository is <https://huggingface.co/datasets/Shanmuk4622/tyre-wear-study>. The reporting snapshot is pinned to revision 22d5a6bc9f953ba3bf2a75919edc7db3193b317b; the HRNet report to a92c0f9c5c1b78c6a06e13d51e18722195230658; and the matched comparison to bbe586c6f00cf12ae4cac8b2e9cb4f875b272abb. The manuscript package records full source paths and SHA-256 digests and recomputes the main geometry aggregates from paired records. Prototype screenshots are local software evidence with recorded hashes, not an invented remote experiment.

Scientific reproduction requires the frozen notebooks, data lineage, manual references, split allocation, and per-run runtime records. The local development environment is not a universal replay lock. Relevant dense-task versions include Ultralytics 8.4.20, Transformers 4.51.3, segmentation-models-pytorch 0.5.0, and timm 1.0.15. Local workstation checks use

PyTorch 2.5.1 with CUDA 12.1 and OpenCV 4.11.0. Different phases have separately recorded numerical environments and should not be presented as one identical runtime.

The paper build uses the provided IEEE Access class and independently regenerated vector plots and numeric tables. It preserves the frozen upstream evidence. Rebuilding the paper is a CPU-only documentation task, distinct from rerunning training or independently replicating the study. Public artifact access does not by itself establish redistribution permission for every source image or external weight; those permissions require author review before submission.

VIII. CONCLUSION

This pilot connects mileage-proxy recognition, manually supervised localisation, matched boundary learning, and an inspectable desktop workflow. On the available collection, classification depends strongly on endpoint and fold; predicted crops and fixed probability fusion do not improve the frozen classifier on average. HRNet reduces matched horizontal boundary error by 23.75% relative to segmentation-derived points on a two-tyre test, with complete comparator coverage and differing supervision. The implemented workstation preserves frame identity, exposes failure states, and supports target-assisted measurement software. These results justify continued component and data validation, while leaving physical wear, external generalisation, and real alignment accuracy unresolved. Stronger evidence now requires better independent measurements and sampling, rather than a larger count of model runs on the same small cohort.

AUTHOR DECLARATIONS FOR REVIEW

Author names, order, affiliation, and corresponding-author details in this draft follow the repository's citation metadata and require confirmation. Funding, competing interests, individual contributions, and image-use/collection permissions must be supplied or confirmed by the authors before submission; no approval, exemption, or absence of funding is asserted here. AI-assisted tools were used to help draft, format, and check this manuscript against the available records. The authors must review the final text, verify the interpretation, and take responsibility for the submitted version.

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